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Geographic accessibility and completion of initial low-dose CT-based lung cancer screening in an urban, safety-net population

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Abstract

Background: Recent modifications to low-dose computed tomography (LDCT)-based lung cancer screening guidelines increase the number of eligible individuals, particularly among racial and ethnic minorities. Because these populations disproportionately live in metropolitan areas, we

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analyzed the association between travel time and initial LDCT completion within an integrated, urban safety-net healthcare system.

Patients and Methods: Using ESRI's StreetMap Premium, OpenStreetMap, and r5r package in R, we determined projected personal vehicle and public transportation travel times between patient residence and the LDCT screening facility for LDCT ordered between March 2017 and December 2022 at Parkland Memorial Hospital in Dallas, Texas. We characterized associations between travel time and LDCT completion in univariable and multivariable analyses. We tested these associations in a simulation of 10,000 permutations of personal vehicle and public transportation distribution.

Results: A total of 2,287 patients were included in the analysis, of whom 1,553 (68%) completed the initial ordered LDCT. Mean age was 63 years; 73% were under-represented minorities (URM). Median travel time from patient residence to the LDCT screening facility was 17 minutes by personal vehicle and 66 minutes by public transportation. There was a small difference in travel time to the LDCT screening facility by public transportation between completed and not completed LDCT (67 versus 65 minutes; P=0.04) but not by personal vehicle (17 minutes for completed and non-completed LDCT; P=0.67). In multivariable analysis, LDCT completion was not associated with projected travel time to the LDCT facility by personal vehicle (OR=1.01; 95% CI, 0.82–1.25) or public transportation (OR=1.14; 95% CI, 0.89–1.44). Similar results were noted across travel type permutations. Black individuals were 29% less likely to complete LDCT screening compared to White individuals.

Conclusion: In a predominantly URM urban population, projected travel time is not associated with initial LDCT completion in an integrated healthcare system. Other reasons for differences in LDCT completion warrant investigation.

Background

Despite a clear mortality benefit,[1] United States Preventive Services Task Force (USPSTF) endorsement, and Centers for Medicare and Medicaid (CMS) coverage, uptake of lung cancer screening with annual low-dose computed tomography (LDCT) remains remarkably limited. As of 2021, it is estimated that only 6% of eligible individuals had undergone LDCT-based screening.[2] This rate lags critically below those for breast cancer and colorectal cancer, which both exceeded 60% in 2021.[3–4] Social, economic, and psychological factors may contribute to the limited uptake of lung cancer screening.[5] Among others, proposed explanations include lack of patient and clinician awareness, associated stigma and fear, and lack of available programs.

In 2021, the USPSTF expanded eligibility for lung cancer screening by lowering the minimum age from 55 to 50 years and the minimum smoking history from 30 to 20 pack-years.[6]

Broadened LDCT eligibility are projected to have major beneficial public health effects, including more cancers detected and more lives saved. The new guidelines will also have an outsized reach to socioeconomically disadvantaged populations, including groups with less health literacy, lower rates of private health insurance, and lower education and income

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levels. In general, these individuals have lower rates of cancer screening participation and adherence.[9–12]

The non-White U.S. population newly reached by revised USPSTF criteria is over-represented in urban areas. Overall, racial and ethnic minorities make up 43% of the nation's urban population, compared to 22% in rural areas.[13] While some rural regions with high smoking rates have a clear shortage of established lung cancer screening programs and LDCT-certified radiology centers, larger cities are more likely to have major medical centers with these resources. Rather than lacking screening sites altogether, the racial and ethnic minorities as well as the socioeconomically disadvantaged individuals living in urban areas instead face challenges accessing these existing programs.[14]

To address this issue central to the success of expanding lung cancer screening programs nationwide and explore a potential barrier to screening adherence for these populations, we analyzed initial LDCT completion according to patient residence and transportation requirements in a major metropolitan area.

Methods

This study was approved by the Institutional Review Board (study #122015–046).

2.1 Study setting and population: This study analyzed patients for whom an initial LDCT-based lung cancer screening was ordered at Parkland Memorial Hospital, which is the flagship location for the integrated safety-net health system for Dallas County (population of 2.6 million; 41% Hispanic, 24% Black). Parkland includes a main campus encompassing a 982-bed tertiary care hospital and specialty clinics, as well as 12 community-based primary care clinics dispersed throughout the county. An enterprise-wide electronic health record (EHR) system (Epic, Verona, WI) allows electronic tracking of a wide array of patient characteristics and outcomes.[15]

Since inception in 2017, lung cancer screening at X facility is generally initiated and overseen by primary care providers (physicians and advanced practice providers), who order LDCT through an EHR order set that incorporates both eligibility assessment (e.g., age and smoking history) and process requirements (e.g., shared decision making). LDCT results are reported to the ordering clinician in template form using the Lung-RADS scoring system. While most primary care medical services in the system occur in the community-based clinics, LDCT for lung cancer screening is available only at the main campus.

For this analysis, our cohort comprised patients for whom initial LDCT-based lung cancer screening was ordered from March 2017 (first availability of LDCT-based lung cancer screening at Parkland) through December 2022. Eligible patients were those for whom we were able to calculate public transit times from their residence to (1) the clinical site at which the first LDCT order was placed (hereafter referred to as "LDCT ordering clinic"), and (2) the LDCT performance site (hereafter referred to as "LDCT screening facility").

2.2 Data Collection: We collected patient residential address at time of initial LDCT order placement, the address for each LDCT ordering clinic, and the address for the LDCT

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screening facility from the EHR. We defined the outcome of our study, initial LDCT screening completion, as “yes” if the patient completed the first LDCT order placed. We did not designate a specific timeframe for completion. Using the EHR, we captured the following patient-level characteristics: age, race and ethnicity, gender, insurance type, and marital status. To measure general health status, we calculated the Charlson comorbidity index (CCI; no significant comorbidity [0–1], average [2–4], severe [≥ 5]) in the year prior to LDCT ordering.

2.3 Data analysis: We geocoded addresses for patient residence, the LDCT ordering clinic, and the LDCT screening facility in ArcGIS Pro 3.0.2 using ESRI Street Map Premium 2022 Q3 as reference data. We then calculated travel time by personal vehicle and public transportation, in minutes, from: (1) patient residence to the LDCT ordering clinic, and (2) patient residence to the LDCT screening facility. We downloaded public transit data in general transit feed specification for X County and two neighboring counties from online repositories. Travel time by public transportation was calculated using the *r5r*package[16] in R with Open Street Map as reference and network data.

To obtain data for a representative weekday and time, we calculated the public transit travel time for all trips starting between 9 am and 11 am on Wednesday, January 25, 2023, and took the median of these theoretical trips as the travel times of interest. We selected a two-hour transit window because a one-hour window provided less stable results due to infrequent components of public transportation. We selected the specific time window to avoid rush hour effects which were not incorporated into our estimates of travel time by vehicle. Similarly, we selected the specific day of the week (Wednesday), as it may be less subject to differential traffic delays than are the beginning or end of the workweek. We limited walking segments to 30 minutes and assumed a walking pace of 3.6 km/hr (~27 min/mi) such that walking segments were limited to ~1.125 miles.

We described patient characteristics and travel times using means (SD) and medians (IQR) for continuous variables and N (percentages) for categorical variables. We mapped mean personal vehicle and public transportation travel times from patient residence to the LDCT ordering clinic and to the LDCT screening facility. We used unadjusted (univariable) and adjusted (multivariable) logistic regressions to evaluate the associations between projected travel time measures (log-transformed) and LDCT screening completion for four different models: (a) patient residence to the LDCT ordering clinic by private vehicle, (b) patient residence to the LDCT screening facility by private vehicle, (c) patient residence to the LDCT ordering clinic by public transportation, and (d) patient residence to the LDCT screening facility by public transportation. We included all covariates in the multivariable model due to their conceptual relevance. We considered statistical significance as $P < 0.05$. We also evaluated the correlation between patient characteristics and travel time measures using Spearman’s rank correlation for continuous variables (e.g., age), Wilcoxon Rank Sum Test/Mann-Whitney Test for sex, and Kruskal-Wallis rank sum test for other categorical characteristics with >2 groups. We conducted all analyses in R.[19]

To account for potential effects of COVID-19 on travel times and general access to healthcare, we conducted a sensitivity analysis to explore whether results varied according to

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COVID period. Because X County pandemic-related shelter-in-place orders began on March 23, 2020, and ended on March 10, 2021, we divided our study into three time periods based on the order date for a patient's first LDCT: (1) pre-pandemic (ordered before February 1, 2020); (2) pandemic restriction period (ordered February 1, 2020, through March 31, 2021), and post-pandemic restriction period (ordered after March 31, 2021). We ran our fully adjusted models with robust standard errors on these three subsets.

Because we were unable to determine whether an individual patient would have taken private vehicle or public transportation to a given scheduled appointment, we conducted a simulation experiment with 10,000 random selections to determine whether our results are robust to a mix of transportation modes (see Supplemental Methods).

Results

Among 3,081 patients for whom an initial LDCT order was placed during the study period, 2,287 (74%) had available public transit times to the LDCT ordering clinic and to the LDCT screening facility. For this cohort, median age was 63 years, 42% were women, 73% were URM, 50% were partnered or previously partnered, and almost all had either moderate or severe comorbidity. Additional demographic characteristics are shown in Table 1. Consistent with the mission of the healthcare system as the safety-net provider for X County, 2,267 patients (99%) resided within X County.

Median street network distance between patient residence and LDCT ordering clinic was 6.7 miles, between patient residence and LDCT screening facility was 10.7 miles. Median projected travel time from patient residence to the LDCT ordering clinic was 12 minutes by personal vehicle and 63 minutes by public transportation. From patient residence to the LDCT screening facility, median travel time was 17 minutes by personal vehicle and 67 minutes by public transportation. Figure 1 displays projected travel times according to trip (patient residence to LDCT ordering clinic; patient residence to LDCT screening facility) and modality (personal vehicle; public transportation).

Overall, 1,553 patients (68%) completed the initial ordered LDCT. Figure 2 displays the proportion of patients completing the initial LDCT according to patient residence location. Univariate associations of patient characteristics and projected travel times with initial LDCT completion are shown in Table 1. Older individuals and those with severe comorbidity were more likely to complete screening than were individuals with moderate or no significant comorbidity. Travel time measures were not associated with LDCT completion. Specifically, median projected travel time by personal vehicle from patient residence to LDCT screening facility was 16.7 and 16.6 minutes, respectively, for patients who completed LDCT and those who did not (P=0.67). Via public transportation, travel time was 67 and 66 minutes respectively for patients who completed LDCT and those who did not (P=0.04).

Multivariable analysis of associations between initial LDCT completion and both patient and travel time characteristics is shown in Table 2. Projected travel times were not significantly associated with LDCT completion. Increasing age was associated with greater likelihood

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of LDCT completion. Black individuals were 29% less likely to complete an initial LDCT compared to White individuals.

Although projected travel times were not associated with initial LDCT completion rates, they correlated with patient characteristics (Table 3). Older age was associated with longer travel times by private vehicle, while females had longer travel times by public transportation. White and unknown race/ethnicity individuals had longer travel times across scenarios. Marital status, comorbidity burden, and insurance type also were correlated with travel times.

Our sensitivity analysis examining the potential effect of COVID-19 on our findings found that travel time coefficient estimates and their significance did not vary among pre-pandemic, pandemic, and post-pandemic periods (Supplemental Table 1). In our simulation experiment of 10,000 random selections of private vehicle/public transportation mix, only 7% of replications for patient residence to LDCT screening facility and 4% of replications for patient residence to LDCT ordering clinic were statistically significant. These had small effect sizes (see Supplemental Results and Supplemental Figure 1).

Finally, because the distribution of travel time measures was skewed, we performed the following: (1) analysis of the longest 10% and 25% of travel times, and (2) analysis of projected travel time measures by quartiles and quintiles as primary independent variables. As shown in Supplemental Table 2 and Supplemental Table 3, we found no significant associations with initial LDCT completion.

Discussion

We studied the association between completion of a patient's first LDCT order and their estimated travel time in a large, urban, safety-net patient population. Although travel times to ordering and screening clinics were associated with numerous patient characteristics —including age, race/ethnicity, comorbidities, marital status, and insurance type—initial LDCT completion was robustly associated only with patient age and race/ethnicity. We found no evidence that longer projected travel times to the LDCT ordering clinics or the LDCT facility are associated with LDCT completion.

Transportation-related challenges represent a potential barrier to accessing healthcare services, particularly for older adults and underserved populations.[5,14,20] Earlier studies examining the association between travel requirements and screening behaviors have demonstrated mixed findings. Aligning with our results, some mammography and colonoscopy studies found no link with travel time,[21–23] while others have revealed an association with uptake.[24] A recent analysis of lung cancer screening in an integrated healthcare system in South Carolina found that longer travel time to the screening site was associated with lower screening utilization.[25] These discordant results may reflect differences in geographic unit under study (county vs census tract), population density (rural vs urban areas), outcome variable (screening initiation vs follow-up), and travel calculations (patient residence vs centroid of zip-code area).

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Providing access to cancer screening programs in urban areas is critical to achieving equity in healthcare. Over 90% of individuals who are Latino live in urban areas (nearly 50% in a central city), compared to less than 70% (20% in a central city) of individuals who are White.[26] Residents who are Black make up only 8% of the rural but 13% of the urban U.S. population.[13] We found that Black persons were less likely to complete LDCT screening compared to White persons in an urban safety-net health care system, consistent with previous studies that have linked lung screening uptake with race and ethnicity.[25,27,28] Among others, potential reasons for this observation include education level, health literacy and beliefs, and smoking status. Notably, a recent study examining travel time to treatment facilities for lung cancer found that a 15-minute increase in driving and public transit time increased the risk of undertreatment and delayed treatment for non-Hispanic Black individuals, but not for non-Hispanic White individuals.[29]

Recent years have seen major growth in the number of programs featuring mobile cancer screening units (mammography, LDCT, Pap tests), not only in rural regions but also in heavily populated urban centers.[30] Yet our results suggest that—in one of the largest, most diverse cities in the U.S.—distance and required time to reach a single, brick-and-mortar screening facility do not seem to affect screening completion rates. The nature of screening referral and performance may underlie these findings. In the healthcare system we studied, LDCT for lung cancer screening could only be ordered by a treating clinician. Accordingly, LDCT order placement implied a previously demonstrated ability to access the healthcare system. Our observed LDCT completion rate of nearly 70% may also indicate specific characteristics of the study setting, including the “closed” hospital system and assurance that screening would be paid for. In considering whether a mobile screening unit is worthwhile for a given healthcare system, it is important to recognize potential values beyond access, such as the increase in general screening and specific program awareness that comes with a region’s citizens seeing mobile units during their everyday activities.

To our knowledge, this is the largest study to examine the association between travel time and lung cancer screening adherence, and the first to incorporate public transportation-based data. This second point is key to studying under-represented populations. Eighteen percent of Black households in the U.S. do not have a car, compared to only five percent of White households.[31] Additionally, we used precise locations of patient residences and clinical facilities rather than census tract centroids as surrogates of proximity, permitting more accurate projections of travel time. Another strength of our study lies in the robustness of our results, as our findings for travel time persisted throughout several distinct approaches for analysis, including accounting for potential COVID effects, focused analysis of outlying travel time values, and examination of associations across 10,000 permutations of transportation type.

This last approach is critical to addressing a limitation inherent to most studies of patient travel—lack of information on the type of transportation a patient used or would use to reach a clinical facility.[32,33] Nearly 1 in 10 urban-dwelling older adults report recently using public transportation, and over 20% of these individuals rely on public transit to see their primary doctor.[34] Although we were unable to incorporate historic traffic information into both measures of travel time, we were able to create comparable measures of travel time.

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Another limitation is the single-center, urban setting, which differs substantially from rural areas that face a unique set of challenges such as a lower number of screening sites and lack of public transportation.[35] Lastly, because the lung cancer screening program under study is relatively new, we examined only the initial ordered LDCT. With effective lung cancer screening requiring longitudinal surveillance over years, future studies will need to analyze not only initial but also subsequent LDCT, as well as other diagnostic tests recommended to follow up abnormal screens.

Potential reasons why certain patients do not complete intended lung cancer screening are complex and multifactorial. Social, psychological, or economic obstacles could play a role. Unmeasured confounding variables or factors such as higher reported perceptions of patient-physician mistrust in non-Hispanic black patients, physician uncertainty regarding the benefit of LDCT-based lung cancer screening, or medical adherence rates among patients of older age could all potentially contribute to the overall rates of screening adherence.[36–38]

In conclusion, we found that travel times to either the ordering or performance facility do not appear to affect LDCT completion in an urban safety-net healthcare system. The near 70% LDCT completion rate overall may reflect the integrated delivery of clinical care, diagnostic procedures, and therapeutic interventions. Nevertheless, LDCT completion rates were lower among individuals who are Black. Examination of other potential barriers to lung cancer screening in such settings is critical to addressing persistently low screening rates in the U.S.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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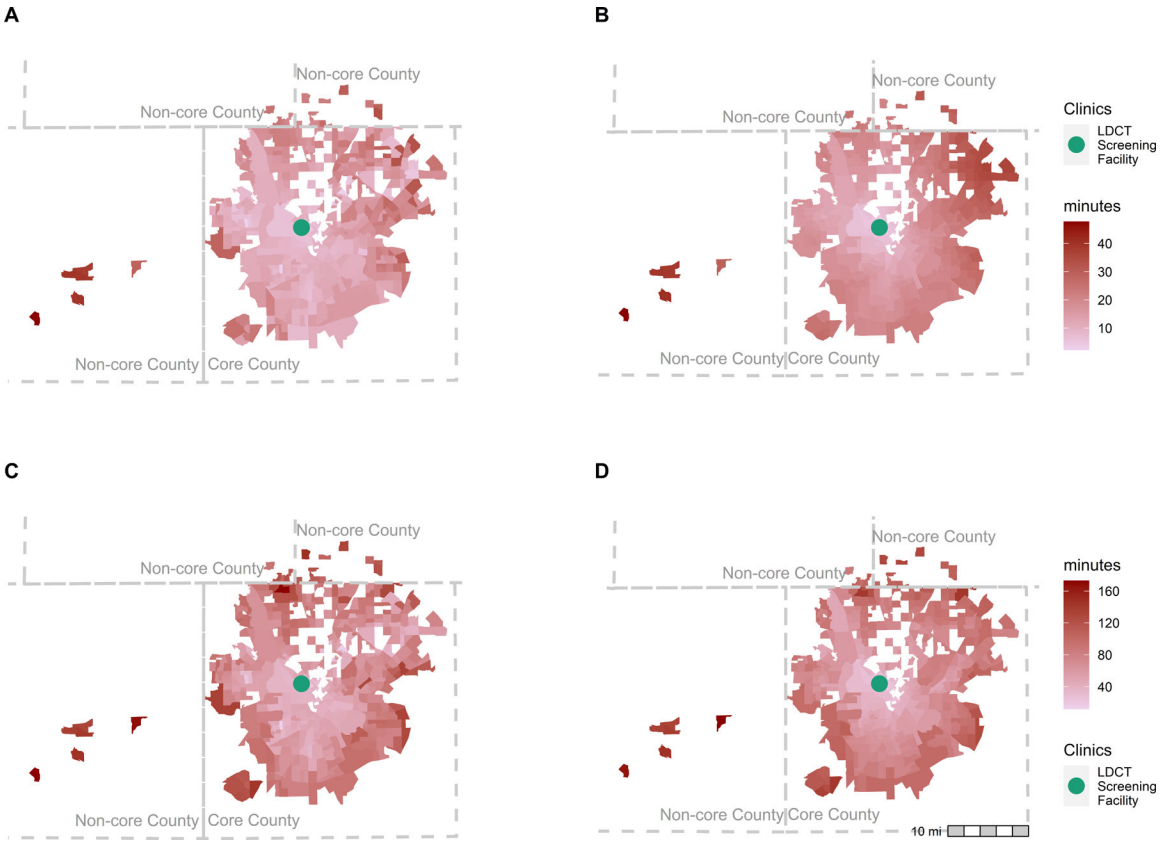


Figure 1.

Distribution of travel times across the study area for *(A)* patient residence to the LDCT ordering clinic by personal vehicle; *(B)* patient residence to the LDCT screening facility by personal vehicle; *(C)* patient residence to the LDCT ordering clinic by public transit; *(D)* patient residence to the LDCT screening facility by public transit.

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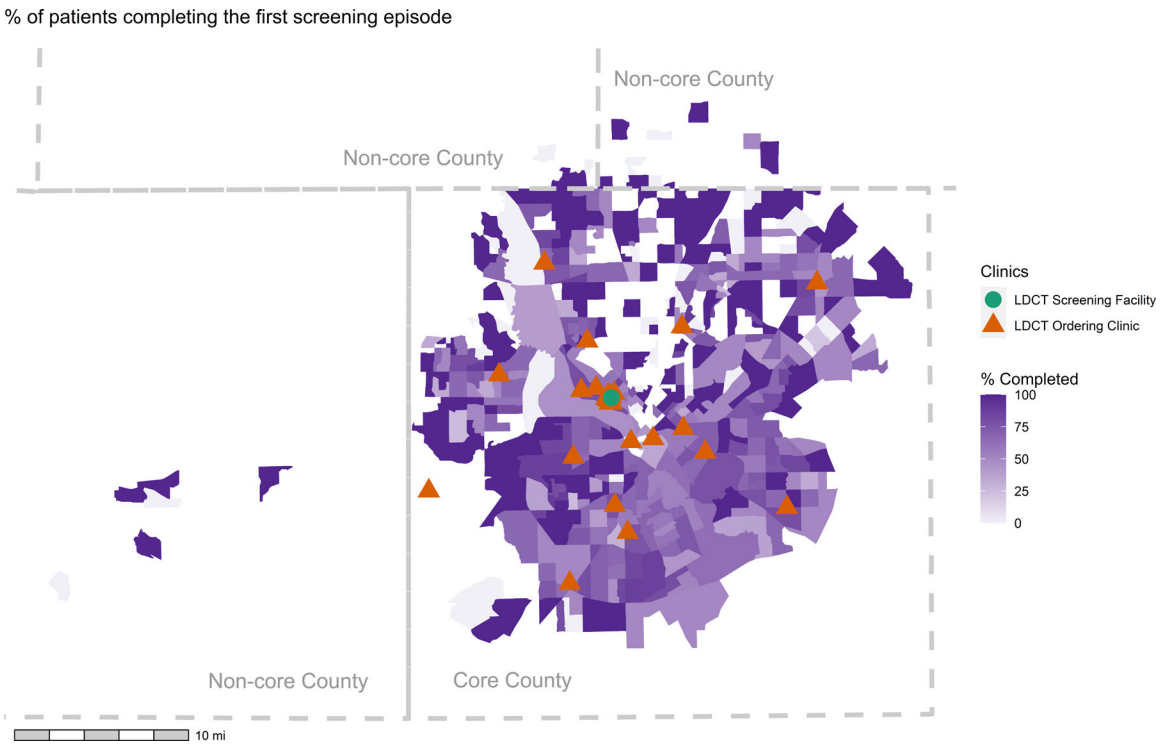


Figure 2.
Distribution of patient residence and proportion of patients completing the initial LDCT screening across the study area.

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Table 1.

Patient and travel time characteristics according to LDCT completion.

	Mean (SD), Median [IQR], or N (%)			
	Not completed (n=734)	**Completed (n=1,553)**	**Total (n=2,287)**	**P-value**
Patient characteristics				
Age in years				<0.0001
Mean (SD)	59.8 (1.2)	64.9 (6.9)	63.2 (6.2)	
Median [IQR]	59.8 [2.1]	64.5 [8.3]	61.5 [7.2]	
Gender				0.70
Male	420 (57)	902 (58)	1322 (58)	
Female	314 (43)	651 (42)	965 (42)	
Race and ethnicity				0.008
White (non-Hispanic)	190 (26)	424 (27)	614 (27)	
Black	426 (58)	808 (52)	1234 (54)	
Hispanic	94 (13)	278 (18)	372 (16)	
Unknown	4 (1)	3 (0)	7 (0)	
Other*	20 (3)	40 (3)	60 (3)	
Marital status				0.50#
Partnered	144 (20)	346 (22)	490 (21)	
Previously partnered	224 (31)	445 (29)	669 (29)	
Single	364 (50)	757 (49)	1121 (49)	
Other or unknown	2 (0)	5 (0)	7 (0)	
Comorbidity index				0.009
No significant comorbidity	8 (1)	12 (1)	20 (1)	
Moderate comorbidity	49 (7)	60 (4)	109 (5)	
Severe comorbidity	677 (92)	1481 (95)	2158 (94)	
Insurance				0.07
Commercial	59 (8)	105 (7)	164 (7)	
Medicare	321 (44)	723 (47)	1044 (46)	

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	Mean (SD), Median [IQR], or N (%)			P-value
	Not completed (n=734)	Completed (n=1,553)	Total (n=2,287)	
Medicaid	178 (24)	314 (20)	492 (22)	
Self-pay or Charity	176 (24)	411 (27)	587 (26)	
Travel time characteristics				
County of residence				0.90
V (Non-core)	3 (0)	9 (1)	12 (1)	
X (Core)	728 (99)	1539 (99)	2267 (99)	
Y (Non-core)	1 (0)	1 (0)	2 (0)	
Z (Non-core)	2 (0)	4 (0)	6 (0)	
Patient residence to LDCT ordering clinic by personal vehicle (minutes)				0.34
Mean (SD)	12.7 (7.2)	13.0 (7.3)	12.9 (7.3)	
Median [IQR]	11.8 [10.2]	12.4 [11.0]	12.2 [10.8]	
Patient residence to LDCT screening facility by personal vehicle (minutes)				0.67
Mean (SD)	16.9 (6.6)	17.1 (6.2)	17.0 (6.3)	
Median [IQR]	16.6 [7.6]	16.7 [6.8]	16.6 [7.1]	
Patient residence to LDCT ordering clinic by public transit (minutes)				0.09
Mean (SD)	63.0 (26.6)	65.1 (27.9)	64.4 (27.5)	
Median [IQR]	61.0 [36.0]	63.0 [39.0]	63.0 [38.0]	
Patient residence to LDCT screening facility by public transit (minutes)				0.04
Mean (SD)	64.6 (21.9)	66.6 (22.1)	66.0 (22.1)	
Median [IQR]	66.0 [30.0]	67.0 [28.0]	67.0 [29.0]	

Table notes: Percentages may not add up to 100 due to rounding.

*: "Other" race and ethnicity includes American Indian, Asian, and other Pacific Islander.

#: Fisher's test p-value

IQR, interquartile range; SD, standard deviation

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Table 2.

Multivariable regression results describing the association between screening completion and patient and travel characteristics.

	Private Vehicle		Public Transit	
	Residence to LDCT ordering clinic	Residence to LDCT screening facility	Residence to LDCT ordering clinic	Residence to LDCT screening facility
Characteristic	**Odds Ratio (95% CI)**	**Odds Ratio (95% CI)**	**Odds Ratio (95% CI)**	**Odds Ratio (95% CI)**
Age (years)	**1.27 (1.24, 1.3)**	**1.27 (1.24, 1.3)**	**1.27 (1.24, 1.3)**	**1.27 (1.24, 1.3)**
Sex				
Male	*ref*	*ref*	*ref*	*ref*
Female	1.16 (0.95, 1.43)	1.17 (0.94, 1.43)	1.16 (0.94, 1.42)	1.15 (0.94, 1.42)
Race/ethnicity				
Non-Hispanic White	*ref*	*ref*	*ref*	*ref*
Black	**0.71 (0.56, 0.89)**	**0.71 (0.56, 0.89)**	**0.71 (0.56, 0.9)**	**0.71 (0.56, 0.89)**
Hispanic	1.04 (0.74, 1.46)	1.04 (0.74, 1.45)	1.04 (0.75, 1.46)	1.03 (0.74, 1.45)
Unknown	0.22 (0.03, 1.42)	0.22 (0.03, 1.41)	0.22 (0.03, 1.44)	0.22 (0.03, 1.47)
Other *	**0.47 (0.24, 0.94)**	**0.47 (0.24, 0.94)**	**0.47 (0.24, 0.93)**	**0.47 (0.24, 0.92)**
Marital status				
Partnered	*ref*	*ref*	*ref*	*ref*
Previously partnered	0.8 (0.59, 1.08)	0.79 (0.59, 1.07)	0.8 (0.59, 1.08)	0.8 (0.59, 1.08)
Single	1.13 (0.86, 1.49)	1.13 (0.86, 1.49)	1.14 (0.86, 1.5)	1.14 (0.87, 1.51)
Other/Unknown	1.32 (0.21, 8.36)	1.3 (0.21, 8.13)	1.37 (0.21, 8.69)	1.35 (0.21, 8.62)
Comorbidity index				
No significant	*ref*	*ref*	*ref*	*ref*
Average	0.76 (0.25, 2.29)	0.77 (0.26, 2.3)	0.76 (0.25, 2.28)	0.76 (0.25, 2.28)
Severe	1.15 (0.41, 3.22)	1.16 (0.42, 3.25)	1.13 (0.4, 3.18)	1.13 (0.4, 3.17)
Insurance				
Commercial	*ref*	*ref*	*ref*	*ref*
Medicare	0.5 (0.34, 0.74)	0.5 (0.34, 0.73)	0.5 (0.34, 0.75)	0.5 (0.34, 0.74)
Medicaid	0.85 (0.57, 1.28)	0.85 (0.57, 1.27)	0.86 (0.57, 1.29)	0.86 (0.57, 1.28)
Self-pay or charity	1.25 (0.83, 1.87)	1.24 (0.83, 1.85)	1.26 (0.84, 1.89)	1.25 (0.83, 1.87)
Log of Travel time (minutes)	1.03 (0.90, 1.18)	1.01 (0.82, 1.25)	1.10 (0.90, 1.35)	1.14 (0.89, 1.44)

Table notes: Odds ratio calculated with logistic regression after adjusting for all listed patient characteristics. Statistically significant OR and 95% CI are represented in bold font.

J Natl Compr Canc Netw. Author manuscript; available in PMC 2025 June 30.

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*: "Other" race and ethnicity includes American Indian, Asian, and other Pacific Islander.

IQR, interquartile range; SD, standard deviation

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Table 3.

Associations between travel time measures and patient characteristics.

	Travel time (min) via Private Vehicle				Travel time (min) via Public Transportation			
	Residence to LDCT ordering clinic		Residence to LDCT screening facility		Residence to LDCT ordering clinic		Residence to LDCT screening facility	
Characteristic	**Mean (SD)**	**P**	**Mean (SD)**	**P**	**Mean (SD)**	**P**	**Mean (SD)**	**P**
Age	**0.04**	**0.04**	0.04	0.08	0.04	0.05	0.03	0.19
Sex		0.58		0.11		**0.04**		**0.004**
Male	12.8 (7.4)		16.86 (6.67)		**63.4 (28.0)**		**64.8 (23.0)**	
Female	12.9 (7.1)		17.23 (5.82)		**65.8 (26.8)**		**67.6 (20.7)**	
Race/Ethnicity		**<0.001**		**0.02**		**<0.001**		**0.002**
NH White	**13.8 (7.6)**		**17.4 (7.2)**		**67.6 (27.5)**		**66.7 (23.5)**	
NH Black	**12.4 (7.0)**		**16.7 (5.8)**		**62.3 (27.3)**		**64.9 (21.3)**	
Hispanic	**12.5 (7.3)**		**17.1 (6.3)**		**64.6 (27.5)**		**67.0 (21.4)**	
Other *	**11.7 (5.4)**		**15.6 (8.8)**		**67.4 (35.8)**		**59.9 (38.8)**	
Unknown	**15.9 (7.4)**		**19.8 (7.0)**		**75.6 (26.3)**		**75.1 (22.7)**	
Marital Status		0.78		**0.03**		0.18		**0.01**
Partnered	12.8 (7.3)		**17.8 (6.2)**		66.0 (27.8)		**68.5 (20.6)**	
Previously partnered	12.9 (7.0)		**17.2 (5.9)**		64.8 (27.2)		**66.9 (20.9)**	
Single	12.9 (7.5)		**16.6 (6.6)**		63.6 (27.6)		**64.3 (23.2)**	
Other/Unknown	10.2 (6.6)		**16.1 (9.4)**		47.6 (22.3)		**60.0 (28.1)**	
Comorbidity Index		**<0.001**		0.28		**0.001**		0.16
No significant comorbidity	**8.8 (8.3)**		15.3 (8.5)		**50.8 (31)**		58.1 (27.4)	
Average comorbidity	**13.0 (7.3)**		17.1 (6.2)		**64.9 (27.4)**		66.3 (21.9)	
Severe comorbidity	**10.6 (6.9)**		16.2 (7.6)		**57.4 (26.8)**		61.1 (23.8)	
Insurance		**<0.001**		0.38		**0.002**		0.41
Commercial	**13.2 (7.2)**		17.0 (6.3)		**64.6 (26.9)**		65.7 (21.7)	
Medicare	**14.7 (7.7)**		17.5 (6.2)		**70.6 (28.0)**		67.4 (23.1)	
Medicaid	**11.7 (7.1)**		17.2 (6.7)		**62.0 (27.9)**		66.7 (22.4)	
Self-pay or charity	**13.1 (7.3)**		16.7 (6.1)		**65.0 (27.9)**		65.3 (22.1)	

Table notes: P-values calculated with Spearman correlation for Age, Wilcoxon Rank Sum Test/Mann-Whitney Test for Sex, and Kruskal-Wallis rank sum test for all other characteristics. Statistically significant correlations are represented in bold font.

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* : 'Other' race and ethnicity includes American Indian, Asian, and other Pacific Islander.